

GUJARAT TECHNOLOGICAL UNIVERSITY
VISHWAKARMA GOVERNMENT ENGINEERING COLLEGE, CHANDKHEDA
Pattern Recognition
SUBJECT CODE: 3171613
Information Technology

Branch		B.E 7 th Odd Semester Information Technology				
Faculty Name		Prof. V. G. Patel				
Starting Date/End Date		18/07/2022				
		<u>PRACTICAL LIST</u>				
Sr. No.	Content	Date	Marks	Sign	CO	
1	Find probability of coin tossing events for multiple coins. - Accept no. of coins from user. - Accept value of multiple tossing of each coin. - Display occurrence probability of Head and Tail for each coin.				CO1	
2	Find probability of throwing generic object like triangular pyramid, square cube ,hexagonal cube etc. - Accept no. of object - Accept no. of trail for each object - Accept each object trail value - Display occurrence probability of each feature of each object.				CO1	
3	Find join probability and conditional probability. - Accept no. of coins - Accept trial count to toss all coins together (Event e). - Find probability of event e1 in which we get at least on ‘Head’ - Find probability of event e2 in which we get at least on ‘Tail’ - Find joint probability of e1 and e2. - $P(e1 \cap e2) = p(e1) - (e2) - P(e1 \cup e2)$				CO1	
4	Simulation of probability density function. Graph-1: Consider single coin tossing two times. $S=\{H,T\} = \{1,0\}$ ideal case. Mean $\mu = 0.5$ Variance $\sigma^2 = 0.25$ Draw graph of Gaussian distribution for this values. $PDF = f(x) = 1/(2 * \pi * \sigma) * e^{(-1/2 * ((x - \mu) / \sigma) ^ 2)}$ For simulation take $\sigma = \sigma/10 = 0.05$ Graph-2: Consider single coin tossing multiple times. $S=\{H,T,T,H,H,H,T,.....\}$ practical case. Calculate Mean $\mu = \sum x_i P(x_i) = COUNT('H')/Total_Trial$ Variance $\sigma^2 = \sum (x_i - \mu)^2 * P(x_i) = \mu * (1-\mu)$ Draw graph of Gaussian distribution for this values.				CO1	

5	Quadratic Equation Solution Write a python program to find solution of equation like..... $x^2 + 7x + 12 = 0$ $5x + \cos(x) + 5 = 0$ $5x + 7y + 8 = 0, 4x + 8y + z = 0 \text{ and } 5y + 3z = 0$ (Hint : Three variables , so three equation are given).				CO2
6	Mean and Covariance Use below link and download Mean and variance simulation. http://cse20-iiith.vlabs.ac.in/ Unzip simulation directory in your computer. In window startup menu open -> JAVA console/JAVA control panel. Go to the Security tab. Check the "Enable Java content in the browser". Click on the button "Edit Site List". - In the new dialogue box that appears, click on "Add". - Type and add the following url to the exception list : "http://test.virtual-labs.ac.in". - Click on the "OK" button of the dialogue box. - Click on the "OK" button of the Java console. - Click on the .jar file of the experiment. Do as directed in lab. section and prepare experiment report and submit same. Also give feedback on VLAB website and attempt given assignments. ** Generate Circle confine samples for class 1 and 2 using proper mean and covariance matrix. ** Generate Ellipse confine samples in horizontal and vertical direction for class 1 and 2 using proper mean and covariance matrix. ** Generate cross Ellipse confine samples in horizontal and vertical direction for class 1 and 2 using proper mean and covariance matrix. ** Put all above three condition screen shots in report.				CO2
7	Bayesian Classification Instructions Use below link and download Bayesian Classification simulation. http://cse20-iiith.vlabs.ac.in/ Unzip simulation dircetory in your computer. In window stratup menu open -> JAVA console/JAVAcontrol panel. Go to the Security tab. Check the "Enable Java content in the browser". Click on the button "Edit Site List". - In the new dilogue box that appears, click on "Add".				CO2

	- Type and add the following url to the exception list : "http://test.virtual-labs.ac.in". - Click on the "OK" button of the dialogue box. - Click on the "OK" button of the Java console. - Click on the .jar file of the experiment. Do as directed in lab. section and prepare experiment report and submit same. Also give feedback on VLAB website and attempt given assignments. ** Generate Circle boundary for class 1 and 2 using proper mean and covariance matrix. ** Generate Ellipse boundary in horizontal and vertical direction for class 1 and 2 using proper mean and covariance matrix. ** Generate cross Ellipse boundary in horizontal and vertical direction for class 1 and 2 using proper mean and covariance matrix. ** Put all above three condition screen shots in report. ** Also generate marks all boundary of classification in each three case.				
8	Decision Boundary Class -1 sample points are [(2,6),(3,4),(3,8),(4,6)] Class -2 sample points are [(3,0),(1,-2),(3,-4),(5,-2)] Find decision boundary in two dimensional space.				CO2
9	Feature Representation. Use below link and download Feature Representation simulation. http://cse20-iiith.vlabs.ac.in/ Unzip simulation directory in your computer. In window startup menu open -> JAVA console/JAVAcontrol panel. Go to the Security tab. Check the "Enable Java content in the browser". Click on the button "Edit Site List". - In the new dialogue box that appears, click on "Add". - Type and add the following url to the exception list : "http://test.virtual-labs.ac.in". - Click on the "OK" button of the dialogue box. - Click on the "OK" button of the Java console. - Click on the .jar file of the experiment. Do as directed in lab. section and prepare experiment report and submit same. Also give feedback on VLAB website and attempt given assignments.				CO3
10	Eigen Vectors and Eigen Values Write a python program to find Eigen Vectors and Eigen Values of given Matrix of 2×2 , 3×3 and 5×5 size. Accept value n from user and Generate $n \times n$ Matrix by random number generator and find Eigen Vectors and Eigen Values of this				CO3

	n * n Matrix.				
11	Handwritten Character Recognition Write a python program to create Artificial Neural Network (ANN) with 2- hidden layer and 1 output layer. Train and test ANN with MNIST dataset of 60,000 hand written digit images. Check accuracy and loss for 2,4 and 8 epoch. Also check accuracy for 1 and 0 (zero) hidden layer.				CO4
12	Handwritten Character Recognition using CNN Write a python program to create Artificial Neural Network (ANN) with 2- hidden layer and 1 output layer. Train and test ANN with MNIST dataset of 60,000 hand written digit images. Check accuracy and loss for 2, 4 and 8 epoch. Also check accuracy for 1 and 0 (zero) hidden layer.				CO4